

# Real-Time Indian Sign Language Recognition & Multilingual Sign Generation

Ahmed Aslam M

*B.S.Abdur Rahman Crescent Institute of  
Science and Technology  
Chennai,India  
2828aslam@gmail.com*

Mohamed Aabidh A

*B.S.Abdur Rahman Crescent Institute of  
Science and Technology  
Chennai,India  
mohamedaabidh414@gmail.com*

N.Sabiyath Fatima

*B.S.Abdur Rahman Crescent Institute of  
Science and Technology  
Chennai,India  
✉sabiyathfathima@crescent.education*

**Abstract**— This study introduces an AI-driven Indian Sign Language (ISL) system aimed at bridging the communication gap between the deaf community and non-signers. In India, there is a significant lack of accessible tools that support ISL, making it difficult for hearing-impaired individuals to engage in everyday interactions. To address this, this framework employs deep learning and computer vision techniques for real-time gesture recognition and text-to-sign translation. A Long Short-Term Memory (LSTM) model is used to accurately interpret complex hand movements and facial expressions, which are critical components of ISL. The system also features 3D animated avatars to visually translate spoken or written language into ISL, supporting multiple languages. By enabling effective two-way communication, this solution empowers the deaf community, enhances inclusivity, and promotes digital accessibility. Experimental evaluations demonstrate the system's high accuracy and real-time performance, making it a valuable tool for education, customer service, and daily communication.

**Keywords**— *Indian Sign Language (ISL), AI-driven Sign Recognition, Deep Learning, Computer Vision, Text-to-Sign Translation, 3D Animated Avatars.*

## I. INTRODUCTION

In recent years, advancements in artificial intelligence and machine learning have significantly improved human-computer interaction, leading to the development of intelligent systems that enhance communication accessibility. One of the most crucial areas where AI-driven solutions are making a difference is in assistive technologies for individuals with hearing and speech impairments. Sign languages serve as the primary mode of communication for the deaf and hard-of-hearing communities, but barriers still exist in bridging communication gaps between sign language users and non-signers.

Developing automated systems capable of recognizing and generating sign language can greatly enhance inclusivity and accessibility. However, challenges such as complex hand gestures, variations in signing styles, and real-time processing constraints make the task highly demanding. Effective solutions require robust machine learning models, efficient gesture recognition techniques, and real-time responsiveness to ensure accurate interpretation and generation of sign language.

One promising development in this area comes from Florida Atlantic University, where engineers have created an AI-powered system that utilizes object detection and

landmark tracking to interpret sign language in real-time. [1]. This system represents a major step forward in developing responsive and accurate recognition technologies that can function in practical settings

In addition to these innovations, ongoing research in the field has identified gaps that still need to be addressed. A recent survey highlights the lack of flexible, adaptive machine learning models capable of handling the diverse range of gestures and expressions found in sign languages worldwide [2]. Companies such as Biz4Group have explored advanced approaches like synthetic data generation and 3D modeling to improve recognition accuracy and training data quality [3]. These innovations aim to bridge existing limitations related to hand tracking, spatial awareness, and gesture complexity.

Moreover, the integration of AI in consumer-facing tools is already improving accessibility. For example, Google's Live Transcribe app helps users with hearing impairments by converting speech to text in real time, enabling smoother conversations and interactions in daily life [4]. With continuous advancements in artificial intelligence, there is immense potential to refine AI-based communication tools further, making sign language translation systems more accurate, scalable, and user-friendly. Overcoming current limitations will pave the way for seamless integration of sign language recognition into everyday applications, fostering better communication and inclusivity.

## II. RELATED WORKS

Shin et al. [5] introduced a two-stream feature extraction approach for Korean Sign Language (KSL) alphabet recognition, integrating handcrafted joint skeleton-based features with deep learning-based ResNet101 pixel features. A new KSL alphabet dataset was created, including 31 characters with over 111,600 image samples. The hybrid approach achieved 92.03% accuracy, surpassing existing methods. The study advances vision-based sign language recognition, reducing reliance on expensive sensor-based systems.

Areeb et al. [6] proposed a deep learning system for recognizing emergency-related gestures in Indian Sign Language (ISL). Three models were developed: a 3D CNN, a VGG-16 + LSTM hybrid, and a YOLOv5-based object detection model. Tested on an eight-class video dataset, the YOLOv5 model achieved a mean average precision of 99.6%, significantly outperforming prior approaches. This study emphasizes the importance of leveraging advanced object

detection methods for real-time emergency gesture recognition, addressing critical communication gaps for hearing-impaired individuals.

Sosa-Jiménez et al. [7] developed a bidirectional translator system for Mexican Sign Language (MSL) to enhance communication in primary care. Using a Microsoft Kinect sensor and Hidden Markov Models (HMMs), the system achieved 99% accuracy and 88% F1 score in recognizing 82 signs from medical and general vocabulary. The tool supports real-time MSL synthesis via a signing avatar and is cost-effective, promoting inclusion for deaf patients in healthcare.

Natarajan et al. [8] proposed a hybrid deep neural architecture (H-DNA) for sign language recognition, translation, and video generation. Using MediaPipe and CNN+Bi-LSTM for recognition, and Dynamic GAN for video generation, the model attained over 95% classification accuracy and a BLEU score of 38.06. The framework leverages multilingual datasets and facilitates real-time, bidirectional communication, addressing key challenges in sign language technology.

Rajalakshmi et al. [9] proposed a hybrid deep neural network (hDNN) methodology for sign language recognition, focusing on Indian and Russian gestures. The novel Indo-Russian sign language dataset was created, featuring diverse participants and daily-life vocabulary. The method employs 3D neural networks with atrous convolutions and attention-based Bi-LSTM for spatial and temporal feature extraction, combined with modified autoencoders and hybrid attention modules for discriminative learning. Achieving 99.87% validation accuracy, the framework outperforms state-of-the-art models, offering scalable and sensor-independent solutions for real-world applications.

Kothadiya et al. [10] proposed SIGNFORMER, a Vision Transformer-based encoder model for recognizing static Indian Sign Language gestures. The dataset included 36 classes of alphabet and numeric gestures from publicly available Indian sign language collections. The methodology applied positional embedding and multi-head self-attention mechanisms, achieving 99.29% accuracy within five epochs. The study highlights its potential in human-computer interaction and communication for hearing-impaired individuals.

Shanableh et al. [11] developed a two-stage deep learning approach for recognizing continuous Arabic sign language sentences using a dataset comprising 40 sentences formed from 80 words. The method combines word count prediction and motion image-based segmentation. Transfer learning with Inception-v3 and biLSTM layers was employed. The system achieved a word recognition rate of 97.3% and a sentence recognition rate of 92.6%, outperforming previous models by 1.8% and 9.1%, respectively. This approach advances signer-dependent recognition accuracy for real-world applications.

Faisal et al. [12] introduced the Saudi Deaf Companion System (SDCS), which integrates sign recognition, speech-to-text synthesis, and avatar-based sign generation modules. Leveraging the KSU-SSL dataset with 293 Saudi signs, the system achieved efficient recognition and real-time translation for diverse domains such as healthcare and education. This facilitates bidirectional communication and supports societal inclusion of individuals with hearing impairments.

Zhang et al. [13] proposed SeeSign, a multimodal framework for sign language recognition that integrates surface electromyography (sEMG) and inertial measurement unit (IMU) signals using statistical and contrastive attention mechanisms. The dataset includes Chinese sign language data captured through wearable armbands. SeeSign achieved 93.17% accuracy on isolated words and robust performance on one- and two-handed sentence datasets, highlighting its potential for improving communication for hearing-impaired individuals. The study advances sign language recognition with superior accuracy and robustness.

Al Abdullah et al. [14] conducted a systematic review of sign language recognition, analyzing 58 influential papers up to 2023. The study emphasized deep learning techniques such as convolutional and recurrent neural networks, which demonstrated high recognition accuracy. Integrating non-manual features like facial expressions enhanced performance. The review highlights advancements, challenges, and the potential for real-world applications to improve communication for hearing-impaired individuals.

Gogoi et al. [15] developed an end-to-end Indian Sign Language (ISL) sentence translation method using the Sign Transformer Network. They utilized two datasets, ISL CSLRT with 1036 images over 700 videos and INCLUDE50 with 4287 videos over 263 words, and performed extensive data augmentation for training. EfficientNetB7 extracted features from images, and the transformer model achieved a peak BLEU-4 score of 61.59 and ROUGE score of 71.29, demonstrating potential for accurate ISL sentence translation even with limited datasets. This work significantly advances ISL translation by focusing on sentence-level tasks and leveraging transformer networks.

Najib et al. [16] provided a survey of sign language recognition and translation methods using machine learning and AI tools. Key contributions include a comprehensive review of systems employing hand gestures, facial expressions, lip reading, and speech conversion techniques. Datasets varied across multiple sign languages like ASL and ISL. Methods such as CNN, HMM, and few-shot learning were used, achieving accuracies up to 99.95%. The study highlighted the significance of integrating various functionalities into real-world applications for improved communication support for deaf-mute individuals.

Maruyama et al. [17] proposed a Multi-Stream Neural Network (MSNN) for word-level sign language recognition using datasets such as WLASL and MS-ASL. The MSNN integrates base, local image, and skeleton streams to capture global body movements, facial expressions, and skeletal joint positions. Using I3D and ST-GCN architectures, the method improved Top-1 accuracy by 10–15% compared to prior methods. This study highlights the value of integrating multimodal features for more accurate gesture analysis.

Miah et al. [18] introduced a two-stream multistage Graph Convolutional Network with Attention and Residual connection (GCAR) for sign language recognition. Tested on large-scale datasets like WLASL and MSL, the model achieved accuracies of 90.31%, 94.10%, 99.75%, and 34.41% across different datasets. This method enhances spatial-temporal feature extraction and short-range dependency learning, setting a new standard for scalable and accurate sign language recognition.

Ihsan et al. [19] developed an attention-enhanced CNN-BiLSTM model for word-level medical sign classification to improve patient-doctor interaction among the hearing impaired. Using a novel dataset of 3,596 videos from 20 signers performing 30 medical signs, the system employed MobileNetV2 for feature extraction and BiLSTM for temporal information. Achieving 95.83% accuracy and an F1 score of 93%, it outperformed existing medical sign recognition models by 5%, emphasizing inclusivity in healthcare applications.

Almaazmi et al. [20] developed a deep learning model combining motion images with a Bi-LSTM layer for Arabic Sign Language recognition. Using the ArabSign dataset (93 words, 50 sentences), the model achieved 99.8% accuracy in signer-dependent and 75.3% in signer-independent modes, highlighting strong performance in cross-context recognition.

Özcan et al. [21] proposed a Zero-Shot Sign Language Recognition (ZSSLR) framework using hand and pose features. Their model integrated ResNeXt, MViT2, ST-GCN, and CLIP to align visual and semantic data. It achieved 78.4% top-1 accuracy on ASL-Text and 74.6% on MS-ZSSLR-W/C, demonstrating effective generalization to unseen signs.

Miah et al. [22] introduced GmTC, a hybrid model combining Graph Convolutional Networks, Multi-Head Self-Attention, and CNNs for Multi-Cultural Sign Language (McSL) recognition. Tested on ASL, KSL, BSL, and JSL datasets, it achieved 89.5%–94.2% accuracy, showing robust performance across languages and regions.

Biswas et al. [23] proposed a hybrid CNN-LSTM model with an Inception module and attention mechanism for gesture recognition. Evaluated on Jochen-Triesch ASL, SMSHP, and custom ISL datasets, it achieved 99.77%, 99.23%, and 97.67% accuracy respectively, enhancing human–robot interaction systems.

Jaiswal et al. [24] implemented a Tiny YOLOv2-based hardware accelerator for Hindi ISL recognition using 6548 images across 13 categories. Utilizing Vitis AI and FINN for optimization, it achieved 32.7 FPS, 61.2% mAP, and efficient power usage, supporting real-time embedded sign recognition applications.

### III. DATASET DESCRIPTION

The Indian Sign Language (ISL) is a sophisticated visual language used by the deaf and hard-of-hearing community in India. It communicates meaning through a harmonious blend of hand gestures, facial expressions, and body movements. Unlike spoken languages that rely on auditory and verbal elements, ISL operates through a visual-spatial modality. This means that comprehension and expression in ISL depend heavily on precise handshapes, orientation of palms, movement trajectories, facial muscle cues, and occasionally body posture. Moreover, ISL often requires simultaneous gestures using both hands and subtle temporal variations that convey grammatical structures, emotions, and intensity levels. These complexities pose a unique challenge to automated systems attempting to recognize and interpret ISL gestures in real time.

To capture the rich temporal and spatial characteristics of ISL, A custom dataset comprising 16 unique sign actions was

developed. Each sign was recorded in 30 different sequences, with each sequence containing 30 continuous frames, allowing the model to observe gesture progression over time. The data was collected using webcam feeds of volunteers performing each gesture. This method ensured that the data reflected natural, spontaneous, and user-authentic signing styles, rather than scripted or robotic movements.

To enhance the generalizability and real-world applicability of the system, environmental conditions were intentionally varied during data collection to enhance the model's generalizability. This included recording under different lighting setups, backgrounds, and camera angles, simulating a wide range of possible user environments. This approach helps the model become more robust against background noise, inconsistent lighting, and changes in perspective—all of which are common in real-world deployments.

The gesture recognition pipeline is built on MediaPipe Holistic, a real-time framework by Google that accurately tracks the full body, including pose, face, and hand landmarks. As shown in Figure 1, it processes each video frame to extract spatial keypoints, represented as NumPy arrays for efficient integration into machine learning workflows.

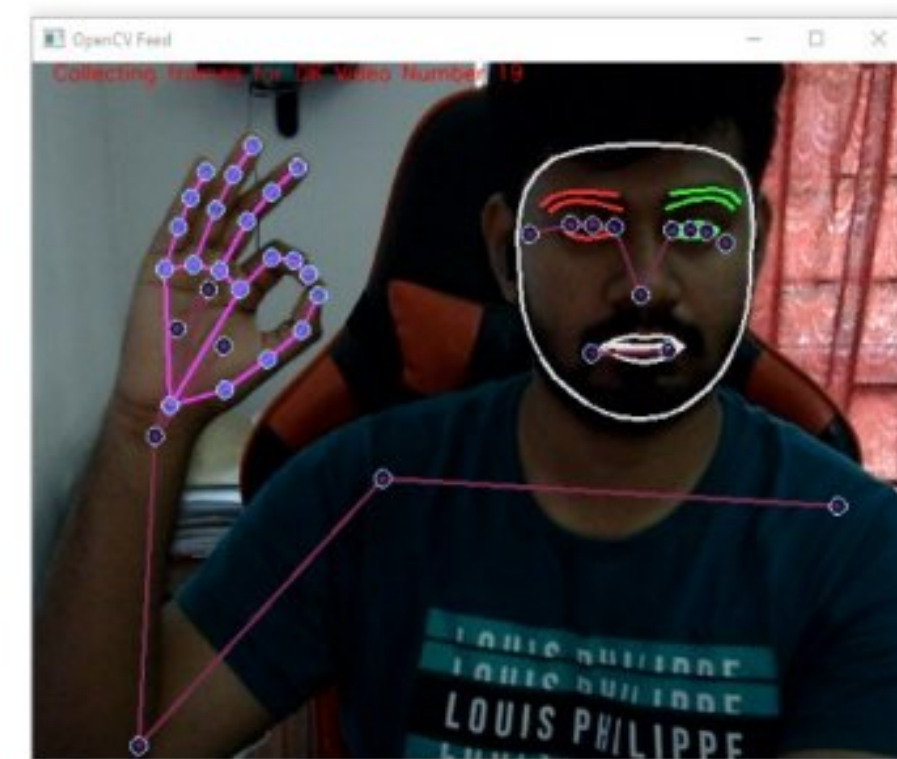


Fig. 1. Keypoint Collection Through Webcam Feed

The resulting dataset was organized into a structured folder hierarchy for clarity and accessibility. As shown in Table I, each gesture (word) has its own dedicated folder, containing the corresponding sequences of landmark data. This systematic structure supports supervised learning tasks, enabling the model to associate specific keypoint sequences with their respective sign meanings.

TABLE I. GESTURE DATASET

| Gesture ID | Gesture Name | No. of Sequences | Frames per Sequence | Total Frames |
|------------|--------------|------------------|---------------------|--------------|
| 0          | angry        | 30               | 30                  | 900          |
| 1          | bring        | 30               | 30                  | 900          |
| 2          | care         | 30               | 30                  | 900          |
| 3          | cry          | 30               | 30                  | 900          |
| 4          | doing        | 30               | 30                  | 900          |
| 5          | food         | 30               | 30                  | 900          |
| 6          | hello        | 30               | 30                  | 900          |
| 7          | help         | 30               | 30                  | 900          |
| 8          | how          | 30               | 30                  | 900          |

|    |         |    |    |     |
|----|---------|----|----|-----|
| 9  | love    | 30 | 30 | 900 |
| 10 | me      | 30 | 30 | 900 |
| 11 | no sign | 30 | 30 | 900 |
| 12 | ok      | 30 | 30 | 900 |
| 13 | sleep   | 30 | 30 | 900 |
| 14 | thanks  | 30 | 30 | 900 |
| 15 | water   | 30 | 30 | 900 |
| 16 | why     | 30 | 30 | 900 |
| 17 | you     | 30 | 30 | 900 |

Overall, the custom ISL dataset, combined with MediaPipe's precise keypoint extraction and varied environmental conditions, provides a solid foundation for developing an accurate and real-time ISL recognition system.

#### IV. PROPOSED METHODOLOGY

The system employs LSTM-based deep learning techniques for sign recognition [25]. Multilingual support includes English, Tamil, and Telugu as mentioned in the Speech to Text Module. The different phases and internal flows within the system are portrayed in Fig. 3, illustrating how various components interact seamlessly. The system works by capturing ISL gestures through a webcam and processing voice inputs via a microphone, ensuring a natural and intuitive interaction. The system recognizes hand movements using deep learning techniques, accurately interpreting gestures in real-time [26]. At the same time, speech is converted into text and then translated into ISL gestures, making it easier for non-signers to communicate with the deaf and hard-of-hearing community.

The proposed LSTM-based model utilizes a sequential deep learning architecture designed to effectively learn and interpret the temporal dynamics of sign language gestures. It begins with an initial Long Short-Term Memory (LSTM) layer consisting of 64 units, which processes input sequences containing the temporal and spatial information of gestures captured through frames, landmark points, and their respective coordinates. The first LSTM layer outputs a sequence of features that is passed to the second LSTM layer, which contains 128 units and is responsible for learning more complex and abstract temporal patterns within the gesture data [27]. Following this, a third LSTM layer with 64 units is used to condense the learned information into a fixed-length feature vector, representing the overall gesture in a compact form. This vector is then passed through a series of fully connected layers with 64 and 32 neurons respectively, which further refine the feature representation. Finally, the output layer, configured with a number of units equal to the total classes of gestures, uses a softmax activation function to perform the

final classification. The model is trained using the Adam optimizer and categorical cross-entropy loss function, both of which are well-suited for handling the multi-class classification challenges inherent in sign language recognition tasks. The dataset is strategically partitioned using a 95:5 train-test split ratio, allocating 95% of the data for model training while reserving 5% for performance evaluation. This division ensures that the model has access to a substantial portion of the available data during the learning phase, while still maintaining a separate test set to assess generalization capabilities across the 18 distinct gesture classes.

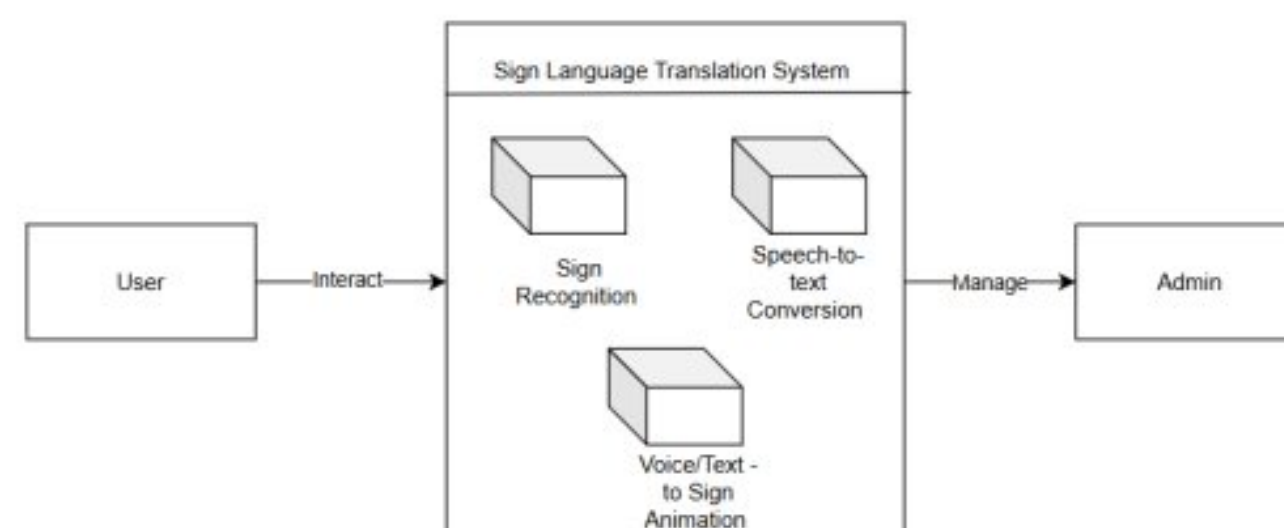


Fig. 3. System Architecture

The sign language interpretation system is structured into two key modules: the User Module and the Admin Module, each serving distinct functions.

##### A. User Module

The User Module, as illustrated in Fig. 4, enables seamless interaction for individuals requiring sign recognition or sign animation. The process begins with user registration, where individuals create an account to access the system's functionalities. Once registered, users can choose between two core features:

- **Sign Recognition:** Users perform signs in front of a webcam, allowing the system to analyze and recognize the gestures. The system processes the visual input and displays the recognized sign as text, facilitating understanding for non-signers.
- **Sign Animation:** Users provide input in the form of speech or text in their preferred language, including English, Tamil, or Telugu. The system then translates the input into the corresponding sign and presents it through animation.

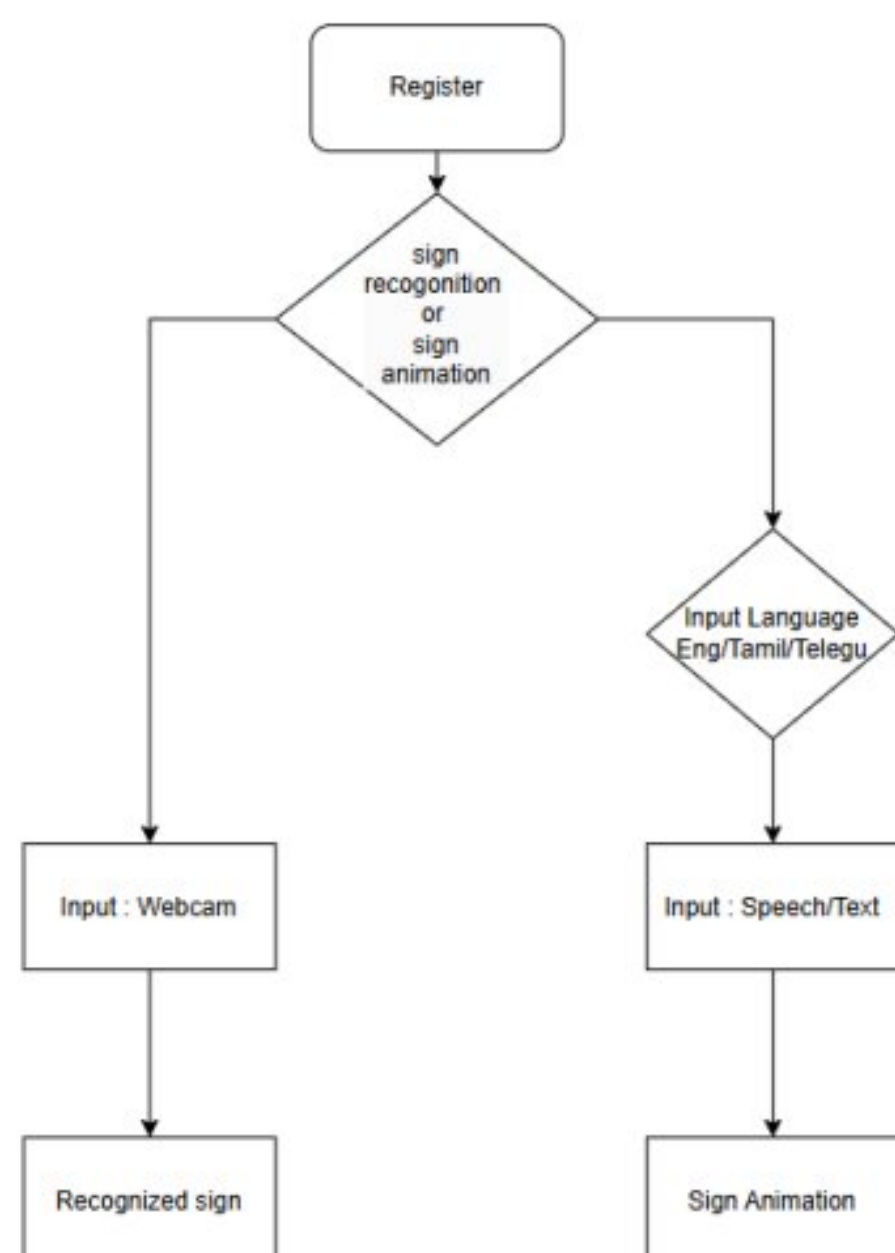


Fig. 4. User Module workflow

### B. Admin Module

The Admin Module, as illustrated in Fig. 5, is designed to manage and maintain the Sign Language Interpretation System by handling sign database operations and model updates. This module ensures that the system remains accurate, up-to-date, and efficient for sign recognition and animation.

The process begins with Admin Login where authorized users, typically developers or backend administrators, access the system. From the backend, they can add, update, or delete sign entries to ensure the database remains current. After making changes, they can initiate model training or updates to improve the system's accuracy in recognizing and animating sign language gestures.

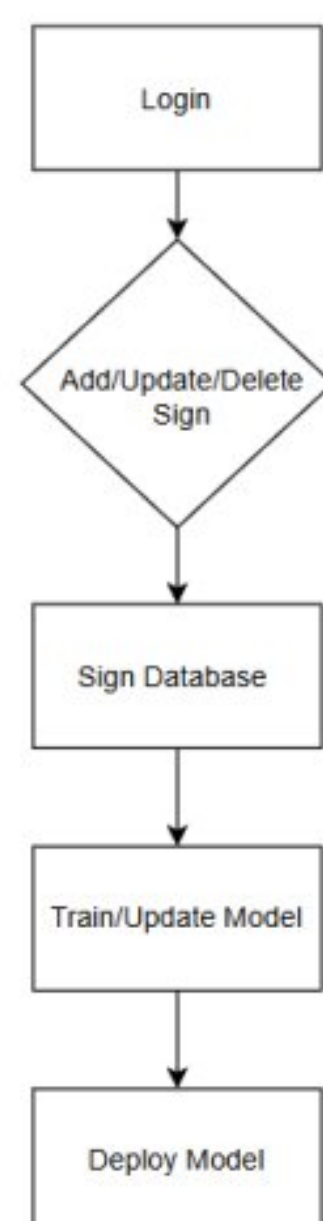


Fig. 5. Admin Module workflow

## V. IMPLEMENTATION

### A. Signed Inference Algorithm for Sign Recognition

The Signed Inference Algorithm plays a crucial role in the Sign Recognition Module by processing captured sign language gestures, mapping them to corresponding labels, and ensuring accurate classification. The algorithm follows a structured approach:

Step 1: Access the stored dataset containing pre-recorded sign gesture sequences.

Step 2: Preprocess and structure data into numerical representations.

Step 3: Train the model on sign sequences to establish base-level constraints for classification.

Step 4: Extract features from input sign gestures using LSTM-based deep learning models.

Step 5: Classify the input gesture based on learned patterns.

Step 6: Refine structured representations into grammatically coherent sentences.

Step 7: Translate the generated English sentence into additional languages, such as Tamil and Telugu.

Step 8: Continuously process new signs, refining the recognition model with real-time updates.

PSEUDOCODE:

START

Assign valid\_sign = 1

Assign detected\_sign = null

Obtain the sign input from the user

Connect to dataset

dataset = load\_data()

model = load\_pretrained\_model()

for each gesture in dataset do

    Extract features from gesture

    Process the sequence using LSTM model

    Evaluate sign classification

    if (recognized\_sign exists in predefined actions) then

        Assign detected\_sign = recognized\_sign

    else

        Implement misclassification handling

        Assign valid\_sign = 0

        return "Unrecognized Gesture"

    end if

Verify consistency with predefined labels

if (sign\_label != expected\_label) then

    Implement correction mechanism

    Assign valid\_sign = 0

```

    return "Incorrect Classification"
end if
Convert ISL gloss to natural English sentence
if (detected_sign is not null) then
    translated_sentence = translate_gloss(detected_sign)
    Translate sentence to Tamil and Telugu
    translated_tamil =
    translate_additional_language(translated_sentence, "ta")
    translated_telugu =
    translate_additional_language(translated_sentence, "te")
    Display translated_sentence to the user
    Display translated_tamil to the user
    Display translated_telugu to the user
end if
Continue evaluating incoming gestures
UNTIL all sequences are classified
if (new input detected) then
    Continue processing the sign recognition
endif
STOP

```

The process flow illustrated in Fig. 6 shows how sign language gestures are translated into English text, beginning with user input and proceeding through model initialization, feature extraction, sign recognition, and translation.

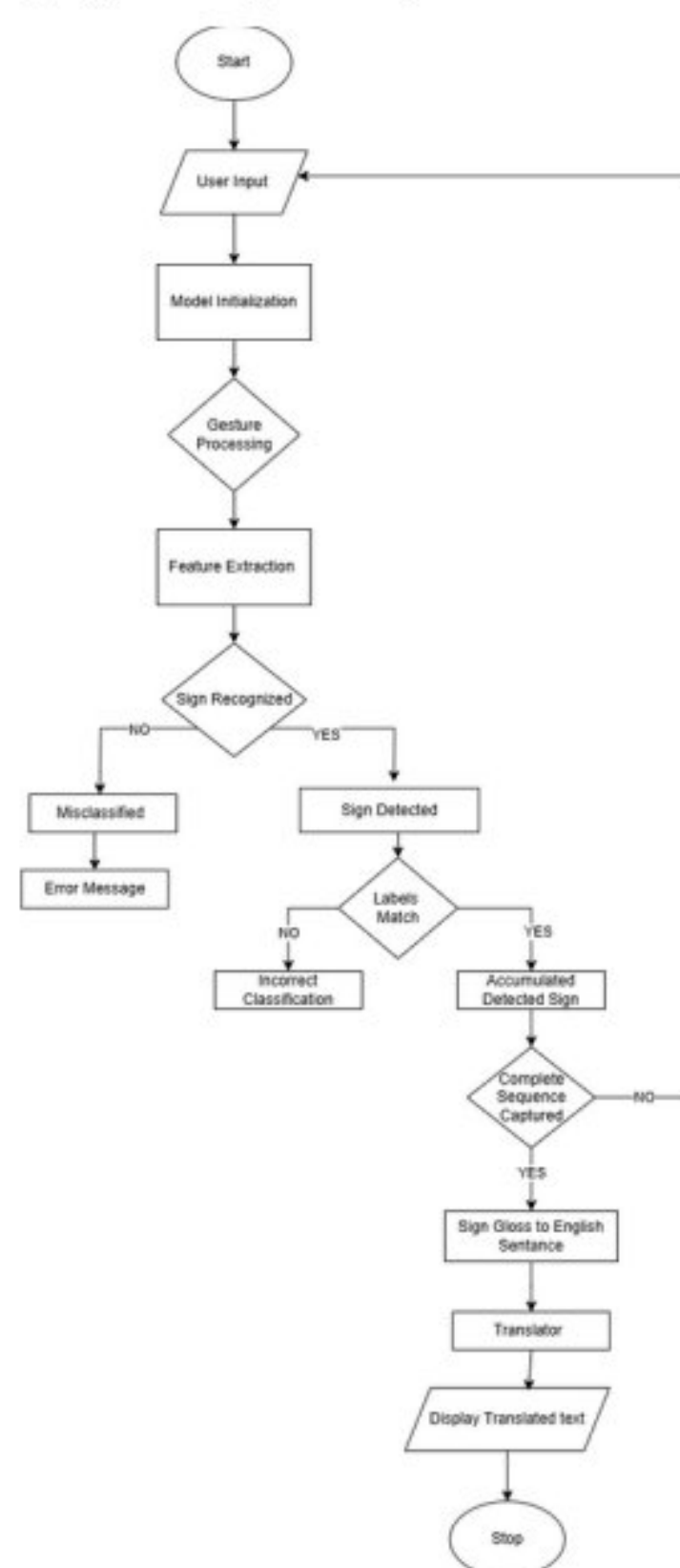


Fig. 6. Signed Inference Algorithm workflow

## B. Speech to Text Module

The Speech to Text Module converts user speech into text as shown in Fig 7 ,through the following steps:

Step 1: The system provides multiple language options for the user to choose from.

Step 2: Once the language is selected, the system connects to the microphone and listens to the user's speech.

Step 3: The captured speech is processed and converted into text using Google's speech recognition API.

Step 4: If the user selected Telugu or Tamil, the system automatically translates the recognized text into English.

Step 5: The transcribed or translated text is displayed to the user.

### PSEUDOCODE:

START

Obtain user's language selection

Access microphone for speech input

Capture audio input

Attempt to recognize speech

IF recognition is successful

Store recognized text

IF selected language is not English

Translate recognized text to English

END IF

ELSE

IF audio is not understood

Display error message

ELSE IF recognition request fails

Display error message

END IF

END IF

Display recognized or translated text

STOP

## C. Text-to-Sign Module

The Text-to-Sign Module translates textual input into sign language animations:

Step 1: The system initializes by setting up an interactive interface and loading a dictionary mapping words to sign language video paths.

Step 2: User input from the speech-to-text module (in English, Tamil, or Telugu) is converted to a natural English sentence.

Step 3: The input sentence is passed to a language model API that translates it into ISL gloss format.

Step 4: The system maps each gloss word to its corresponding sign language video file using the predefined dictionary.

Step 5: The final gloss output and corresponding sign videos are displayed to the user.

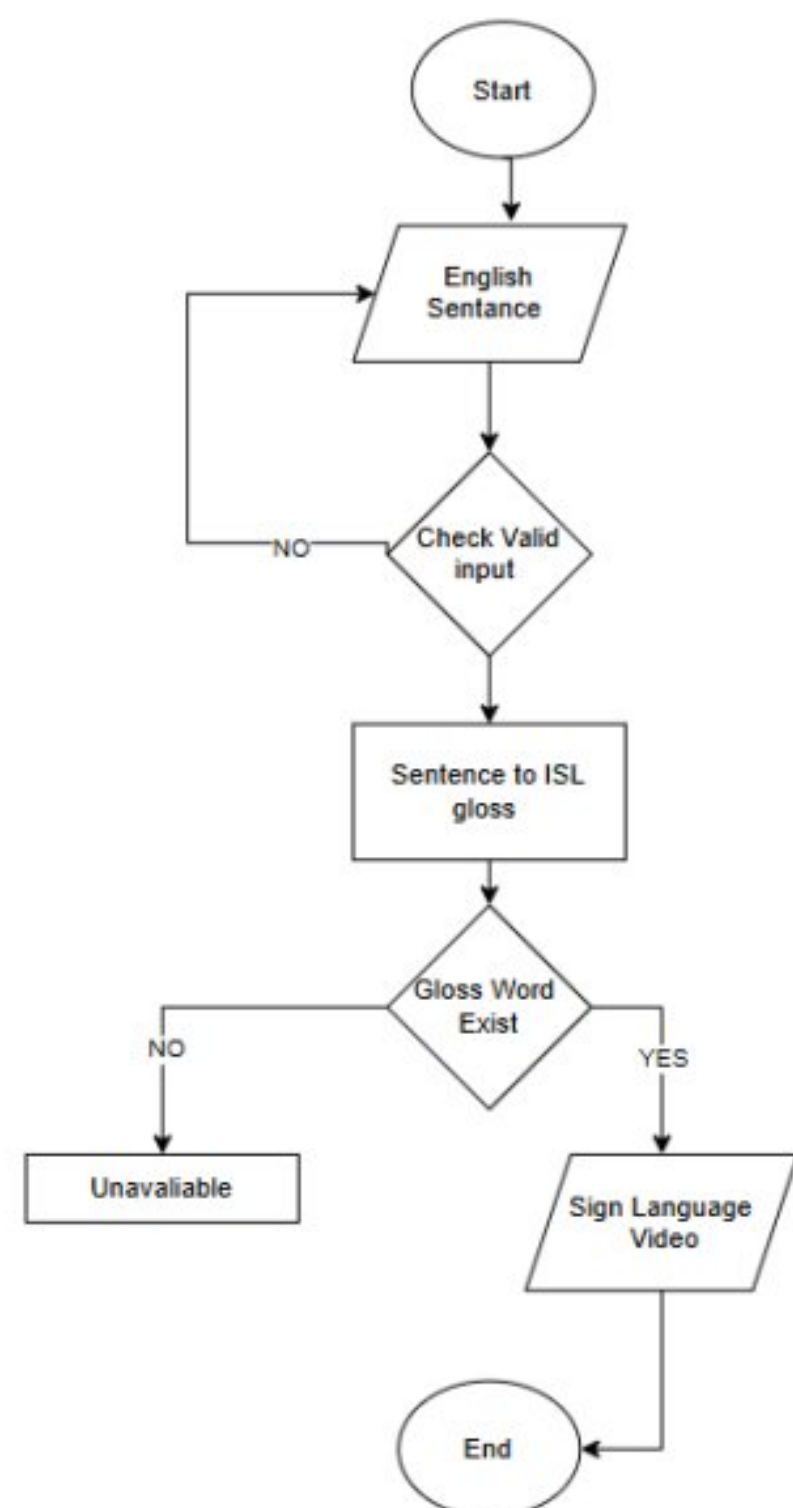


Fig. 7. Text to Sign Module workflow

## VI. RESULTS AND ANALYSIS

Based on the experimental results and visual representations, the proposed sign language recognition system demonstrates significant improvements in accuracy compared to baseline models. Figure 8 illustrates the comparative test accuracy of various deep learning architectures, namely the Proposed LSTM Model, CNN, CNN-LSTM hybrid, Bidirectional LSTM, and Transformer-based models. The results clearly demonstrate the superior performance of the Proposed LSTM Model, which achieves the highest test accuracy of 94.44%. In contrast, the remaining architectures show substantially lower accuracies, ranging between 22.22% and 50%.

This evaluation is based on a dedicated test dataset that was kept completely separate from the training process, ensuring a fair and unbiased assessment of each model's generalization capability. The significant performance gap underscores the effectiveness of the Proposed LSTM Model in learning temporal dependencies and handling sequential patterns crucial for sign language recognition tasks. The results suggest that while CNNs and Transformers may excel in spatial or parallel processing, LSTM-based models are particularly well-suited for modeling temporal dynamics inherent in gesture sequences.

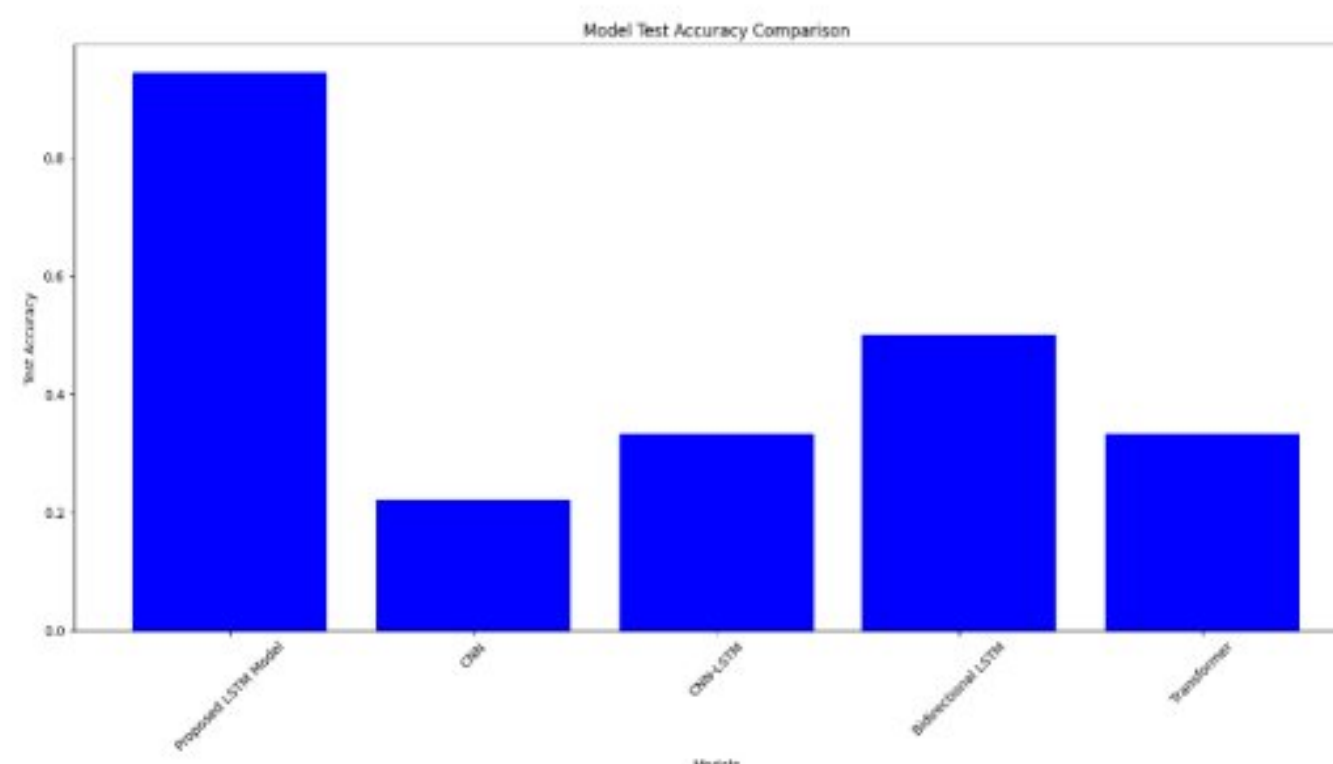


Fig. 8. Test Accuracy Comparison between different architectures

Figure 9, presents the learning curves that depict the changes in training and validation accuracy across successive epochs. These curves represent how well the model classifies data during training and how it performs on a separate validation set not seen during training. Training accuracy is measured using the data the model learns from, while validation accuracy reflects performance on held-out data, providing a measure of generalization. The curves offer valuable insights into the model's learning process, such as how quickly it learns, whether it is improving consistently, and if it begins to overfit or plateau. Stable and converging accuracy curves typically indicate effective training and strong generalization capability. The Proposed LSTM Model consistently achieves higher accuracy rates throughout the training process, while the Bidirectional LSTM model shows unstable learning dynamics. All models were trained using categorical cross-entropy as the loss function and Adam optimizer with default parameters. Early stopping with a patience of 20 epochs was implemented to prevent overfitting, terminating training when validation loss stopped improving. This explains the varying epoch counts visible in the learning curves, as some models converged earlier than others.

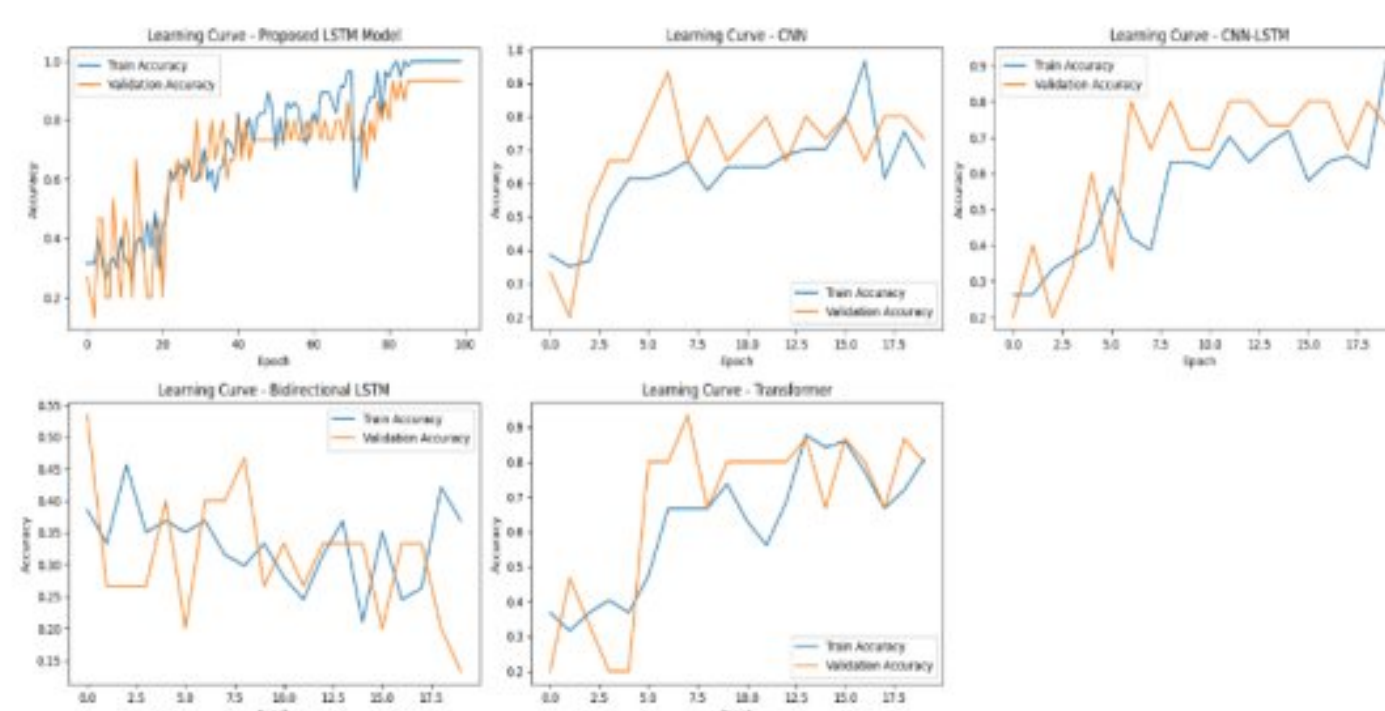


Fig. 9. Comparative study of model performance(Learning curve)

To quantitatively evaluate the performance of the models, we propose the following evaluation formula for the Sign Recognition Module:

$$\text{Accuracy} = \frac{(\text{Total Signs Correctly Recognized}) \times 100}{\text{Total Sign}}$$

Confusion matrices in Fig. 11 further illustrate the classification performance of the proposed LSTM model. With precision (0.94), recall (0.93), and F1 score (0.93), the model demonstrates effective sign language recognition. The classification is achieved for "doing" (100%) and "love" (100% recall), while minor confusions occur between

"hello"/"no sign", "bring"/"care", and "angry"/"cry" (1 instance each). The predominantly diagonal matrix confirms the LSTM model's success in differentiating visually similar signs like "hello" and "thanks" that posed challenges for previous approaches.

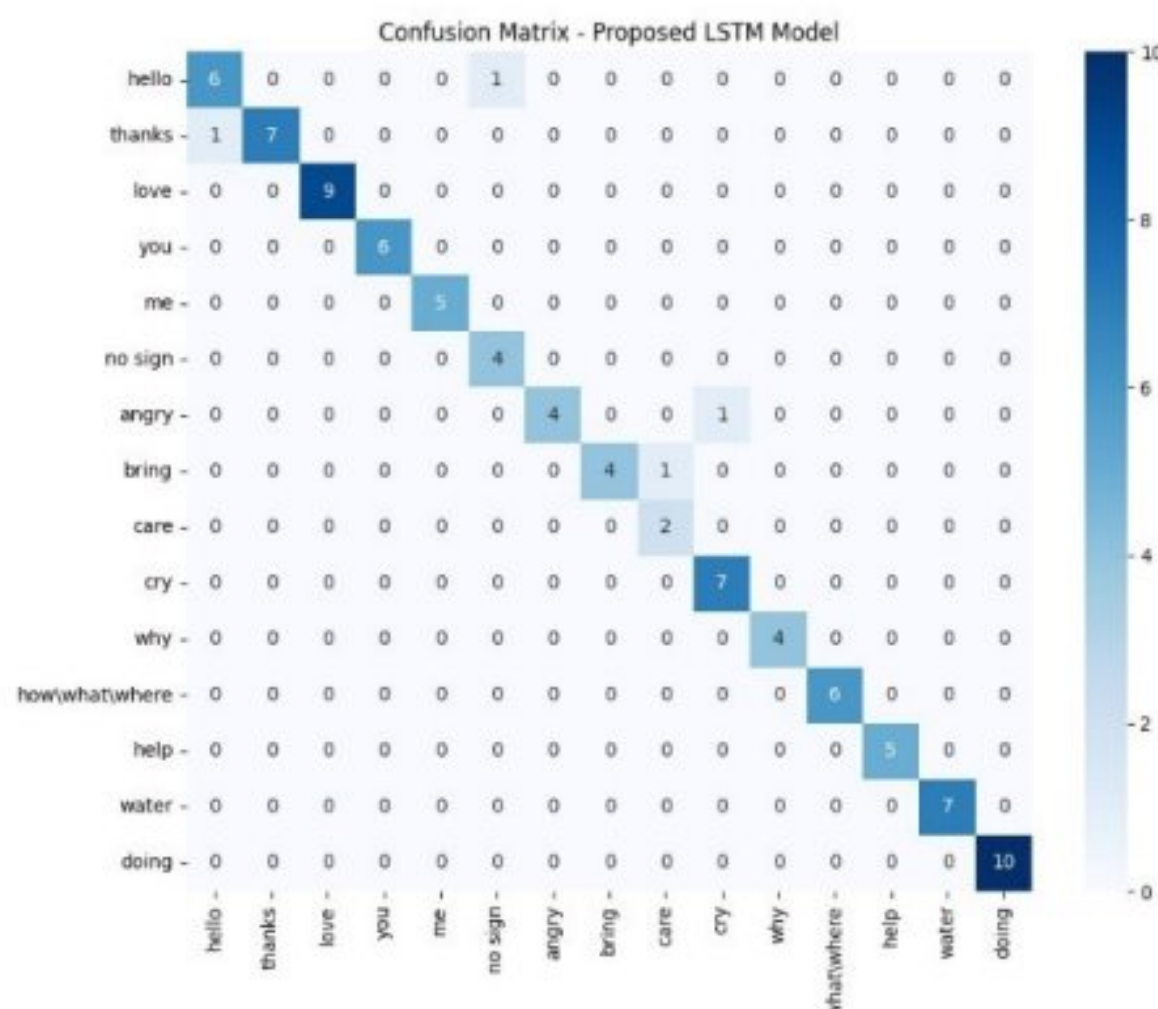


Fig. 10. Proposed Model Confusion Matrix

The analysis confirms that the Proposed LSTM Model excels in recognizing sign language gestures with high precision and stability. Its superior performance can be attributed to its ability to capture sequential dependencies and temporal information.

## VII. CONCLUSION

The proposed AI-driven Indian Sign Language (ISL) recognition and generation system successfully enhances communication accessibility through deep learning and computer vision techniques. By leveraging a Long Short-Term Memory (LSTM) model, trained on a dataset of ISL gestures, the system efficiently recognizes sequential hand movements and facial expressions, ensuring accurate sign detection and text-to-sign translation. The model achieves a high recognition accuracy (94.44%), outperforming baseline architectures such as CNN, CNN-LSTM hybrid, and Transformer-based models. Precision and reliability are achieved by extracting gesture features using LSTM-based deep learning for accurate classification, continuously updating the model with new signs, verifying outputs against predefined labels, and applying correction mechanisms for misclassifications.

To further enhance the system, future work will involve expanding the dataset with more diverse signers and varied environmental conditions. Advanced multimodal learning techniques, combining gesture recognition with facial expressions and lip movements, will be integrated to boost performance. The system will also be optimized for real-time use in mobile and web applications, increasing accessibility for the deaf and hard-of-hearing communities. Additionally, Zero-Shot and Few-Shot Learning methods will be explored to recognize new sign gestures with minimal training data. With continued advancements, this system has the potential to become a comprehensive and scalable ISL translation tool, significantly improving inclusivity and communication for individuals with hearing and speech impairments.

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